FILGRASTIM

Bone pain common

Splenic rupture (rare)

CHEMICAL NAME Recombinant methionyl human granulocyte colony-stimulating factor (r-metHuG-CSF), 175-amino-acid non-glycosylated protein

CLASS Granulocyte colony-stimulating factor (G-CSF)

TRADE NAMES Neupogen · Zarzio · Nivestim (biosimilars) · Neulasta (pegfilgrastim, long-acting)

THERAPEUTIC USES / INDICATIONS

- Prevention of febrile neutropenia after myelosuppressive chemotherapy.
- Mobilisation of peripheral blood stem cells for transplant.
- Severe chronic neutropenia (congenital, cyclic, idiopathic).
- Neutropenia in HIV; acute radiation syndrome.

ADMINISTRATION

- **SC** injection daily (or IV infusion); teach self-injection.
- Start 24–72 h after chemotherapy (not within 24 h before/after).
- Store in fridge; do not shake.
- Pegfilgrastim: single SC dose per cycle (on-body injector option).

DOSAGES

- **Chemotherapy-induced neutropenia:** 5 mcg/kg/day SC until neutrophil recovery (typically 5–10 days).
- **Stem-cell mobilisation:** 10 mcg/kg/day for 5–7 days.
- **Pegfilgrastim:** 6 mg SC once per cycle.

PHARMACOKINETICS

- **Onset:** neutrophils rise within 24 h (transient), sustained in 3–5 days.
- Cleared by neutrophil receptors and kidneys.
- **t½:** 3.5 h (pegfilgrastim 15–80 h).

PHARMACODYNAMICS

- Stimulates **G-CSF receptors** on myeloid progenitors → proliferation, differentiation and release of neutrophils; enhances neutrophil function.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Bone pain (back, pelvis, long bones), myalgia.
- Headache, fatigue.
- Injection-site reactions.

ADVERSE EFFECTS (SERIOUS)

- Splenic rupture (left upper abdominal or shoulder-tip pain).
- ARDS; capillary leak syndrome.
- Sickle cell crisis (in sickle cell disease).
- Glomerulonephritis; aortitis; leucocytosis; possible MDS/AML in chronic use.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to E. coli-derived proteins.
- **Caution:** sickle cell disease, myeloid malignancies.

KEY INTERACTIONS

- **Lithium:** may potentiate neutrophil release.
- **Chemotherapy given within 24 h:** ↑ myelotoxicity.

NURSING MANAGEMENT

- FBC regularly to monitor neutrophil recovery.
- Teach injection technique, storage, sharps disposal.
- Assess bone pain (paracetamol, loratadine may help).
- Report left upper abdominal or shoulder pain urgently.
- Monitor respiratory symptoms.

PATIENT EDUCATION

- Inject at the same time daily as taught; keep in fridge.
- Bone pain is common — take paracetamol as advised.
- Seek urgent help for pain in the upper left abdomen or tip of the shoulder, or breathlessness.
- Still report fever ≥38 °C immediately.

KEY POINTS

- G-CSF; boosts neutrophils.
- Bone pain common.
- 5 mcg/kg/day SC from 24–72 h after chemo.
- Splenic rupture, ARDS (rare).

SOURCES CHECKED **[923]** Neupogen 30 MU (0.3 mg/ml) solution for injection (UK SmPC) **[924]** NEULASTA Pre-Filled Syringe (UK SmPC) **[925]** Releuko, IV, SC (US label) **[926]** Neupogen, IV, SC (US label)

Full references p. 432–457